9.3 ACADEMIC AND RESEARCH INSTITUTES

9.3.1 EXPANDING APPLICATIONS OF MICROBIOME RESEARCH TO DRIVE GROWTH

This segment is inclusive of the application of microbiome sequencing services in environmental and ecological research, veterinary research, and other research activities conducted by academic and research institutes.

Metagenomics is used in microbial ecology and environmental applications to study a wide range of microbial communities present in the physical surroundings (which cannot be cultivated in laboratories). Studying microbes in the environment helps to understand their potential use in various fields, such as agriculture and biotechnology. It also provides valuable insights into other areas, such as the ecology of water bodies, debris filtered from the air, or a sample of soil, to identify the species present in the environment. It can also play a crucial role in identifying a broad range of invasive and endangered species and tracking seasonal populations and variations.

The development of NGS techniques for the large-scale analysis of microbial communities and the rising number of ecological and environmental studies conducted across the globe will support the continued demand for microbiome sequencing in this end-user segment.

Advancements in DNA sequencing methodologies and data analysis have revolutionized research in many areas of veterinary sciences, including medicine, livestock management, and disease management. The veterinary applications of metagenomics include the identification of the complex consortium of bacteria, protozoa, and fungi, among others, and their interaction. The need for rapid and accurate diagnosis of pathogens in livestock species and changes in animal husbandry—including growing herd sizes—are likely to propel the market growth in this segment.

TABLE 77 MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	56.62	64.07	72.45	82.01	92.90	105.62	120.48	138.04	13.8%
Europe	30.81	35.05	39.86	45.38	51.70	59.10	67.81	78.13	14.4%
Asia Pacific	8.52	9.93	11.55	13.44	15.64	18.25	21.36	25.09	16.8%
Latin America	1.95	2.15	2.36	2.60	2.86	3.15	3.48	3.87	10.4%
Middle East and Africa	1.35	1.47	1.59	1.73	1.87	2.03	2.21	2.41	8.6%
Total	**99.25**	**112.66**	**127.82**	**145.15**	**164.97**	**188.15**	**215.34**	**247.53**	**14.1%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 78 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	51.31	58.11	65.77	74.50	84.47	96.10	109.72	125.81	13.9%
Canada	5.31	5.96	6.69	7.51	8.44	9.51	10.76	12.23	12.8%
Total	**56.62**	**64.07**	**72.45**	**82.01**	**92.90**	**105.62**	**120.48**	**138.04**	**13.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 79 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	6.65	7.60	8.68	9.92	11.35	13.03	15.01	17.37	14.9%
UK	5.95	6.83	7.84	8.99	10.33	11.91	13.77	16.00	15.3%
France	4.71	5.36	6.10	6.95	7.93	9.07	10.42	12.02	14.5%
Italy	1.33	1.51	1.71	1.94	2.21	2.52	2.88	3.31	14.1%
Spain	1.02	1.16	1.31	1.48	1.68	1.91	2.18	2.50	13.8%
Rest of Europe	11.14	12.59	14.22	16.08	18.20	20.66	23.55	26.94	13.6%
Total	**30.81**	**35.05**	**39.86**	**45.38**	**51.70**	**59.10**	**67.81**	**78.13**	**14.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 80 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	3.60	4.20	4.89	5.69	6.63	7.75	9.07	10.67	16.9%
Japan	1.68	1.97	2.30	2.69	3.15	3.70	4.35	5.14	17.4%
India	0.21	0.25	0.29	0.33	0.39	0.45	0.53	0.62	16.5%
Rest of Asia Pacific	3.03	3.52	4.07	4.72	5.47	6.36	7.41	8.66	16.3%
Total	**8.52**	**9.93**	**11.55**	**13.44**	**15.64**	**18.25**	**21.36**	**25.09**	**16.8%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 81 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR ACADEMIC AND RESEARCH INSTITUTES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.91	1.01	1.11	1.22	1.35	1.49	1.65	1.83	10.5%
Rest of Latin America	1.03	1.14	1.25	1.37	1.51	1.66	1.84	2.04	10.3%
Total	**1.95**	**2.15**	**2.36**	**2.60**	**2.86**	**3.15**	**3.48**	**3.87**	**10.4%**

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

9.4 OTHER END USERS

This segment is inclusive of the application of microbiome sequencing services in clinical research, diagnostics, and collaborative R&D activities conducted by small-scale CROs, CDMOs, hospitals, and clinics. Biomarkers provided by the human gut microbiome may allow a new means to distinguish between a patient’s physiology and disease conditions. It can also enable mapping microbial fingerprints and defining microbial-derived biomarkers that can help in identifying individuals at high risk of diseases, ensuring the diagnosis of existing conditions, and monitoring the efficacy of treatment.

For example, a bacterial microbiota diversity analysis—used to check for differences in the strain and number of microbes in chronic kidney disease patients and healthy people—found that there was an abundance of Bacteroidetes and Proteobacteria and fewer Firmicutes as compared to a healthy individual. Consequently, this provided a marker for the diagnosis of CKD. Furthermore, there are some diseases that occur because of the translocation of microbes from their normal habitat into some other tissues, where they can trigger diseases by setting up an inflammatory cascade. For example, it was found in a study that lupus, an autoimmune disease, was caused because of the translocation of Enterococcus gallinarum from the gut to the liver and other tissues, which triggered an autoimmune response. These are two instances of the potential use of the human microbiome as a diagnostic tool.

Several firms are focusing on investigation in this area. Enterome SA, for example, raised USD 13.8 million in venture capital funding for developing tests that use the composition of gut bacteria to diagnose inflammatory and liver diseases. It also entered into an agreement with Bristol-Myers Squibb for the discovery and development of microbiome-derived biomarkers, drug targets, and bioactive molecules to be developed as potential companion diagnostics and therapeutics for cancer. This will greatly support the prospects of the human microbiome in clinical research.